

Enea Monzio Compagnoni

Final-year PhD in Computer Science, University of Basel · Zürich, Switzerland

Research Intern at Google (Intrinsic) · Model Optimization & Cross-Platform Edge Deployment

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Summary

Research Scientist and final-year PhD candidate in stochastic optimization and deep learning, currently working at **Google (Intrinsic)** on **model compression, inference optimization, and cross-platform computer vision deployment** for robotic perception. My work spans post-training quantization, quantization-aware training, distillation, graph surgery, accelerator compilation, hardware acceleration, toolchain integration, and on-device benchmarking across **Qualcomm and SiMa.ai accelerator platforms**. I have main-authored seven papers across **ICML, ICLR, AISTATS, and IJCNN** on adaptive optimizers, compressed/distributed learning, differential privacy, and reinforcement learning, with a strong focus on scaling laws. Expected PhD completion in **October 2026**.

Skills

Programming/ML Systems: **Python** (PyTorch), ONNX, Docker, Git, reproducible ML experimentation

Models/Optimization: Computer vision, robotic perception, Vision Transformers, CNNs/ResNets, model training, model compression, PTQ, QAT, distillation, stochastic optimization, RL

Edge AI/Deployment: Qualcomm and SiMa.ai platforms, inference optimization, cross-platform computer vision deployment, hardware acceleration, graph surgery, ONNX export, accelerator compilation, profiling, on-device benchmarking, hardware setup, host-device networking

Mathematics: Stochastic calculus, probability, linear algebra, statistics

Experience

Google -- Intrinsic

Jun 2026 -- Dec 2026

Research Intern, Model Optimization & Cross-Platform Edge Deployment -- Zürich, Switzerland

- Train and optimize Vision Transformer models for robotic perception in PyTorch; benchmark multiple quantization schemes across **PTQ** and **QAT**, for efficient edge inference.
- Build reproducible, end-to-end model optimization and deployment workflows, covering ONNX export, graph surgery, accelerator compilation, runtime integration, and on-device validation.
- Profile model execution and quantization configurations across model accuracy, latency, throughput, memory footprint, numerical fidelity, and accelerator coverage on heterogeneous hardware.
- Evaluate model compression and hardware-acceleration strategies, including quantization and distillation, to improve deployment efficiency under edge-device compute and memory constraints.
- Set up and debug development hardware, SDK and compiler environments, containerized workflows, and host-device networking across both platforms.
- Coordinate multiple internal teams and serve as a technical point of contact for external hardware vendors, clarifying platform constraints, resolving integration issues, and unblocking deployment.

University of Basel

Oct 2022 -- Oct 2026

PhD Researcher & Lecturer, Computer Science -- Basel, Switzerland

- Derived and validated **batch-size scaling rules** for learning rate, AdamW momentum parameters (β_1, β_2), and decoupled weight decay.
- Built reproducible training pipelines in PyTorch for LLMs, ViTs, and ResNets.
- Worked on Differential Privacy optimization and LLM training pipelines under high-privacy regimes.

Flexion Robotics AG

May 2025 -- Aug 2025

AI Engineer -- Robotics Intern (Perception) -- Zürich, Switzerland

- Implemented **VLM/VLA**-based agentic frameworks for autonomous robot operations.
- Developed prompting strategies; integrated components for reliable sim-to-real operation.
- Designed experimental setups for **stable, reproducible, and goal-oriented workflows**.
- Optimized the agent for practical use and coordinated stakeholders toward a **successful real-world robotics demonstration**.

Yahoo Research

Jul 2023 -- Sep 2023

Intern, Scalable Machine Learning -- Munich, Germany

- Built a risk-aware controller for ad-budget pacing; improved spend control by $\sim 4\%$ in A/B tests and delivered reusable Python components for downstream production integration.

UBS AG

Jul 2019 -- Sep 2022

AI Quantitative Analyst -- Zürich, Switzerland

- Designed a Lombard lending risk model for UHNW clients; improved decisioning quality.
- Increased **liquidity-forecasting** accuracy for the Japanese market by $\sim 3\%$; enhanced AML detection with new features and ML pipelines.

Education

University of Basel

Oct 2022 -- Oct 2026

PhD in Computer Science

ETH Zürich

Sep 2018 -- Apr 2022

MSc in Quantitative Finance (GPA 5.76/6, Summa Cum Laude)

University of Milan

2012 -- 2018

MSc & BSc in Mathematics (110/110, Summa Cum Laude)

Publications

First-Author Publications

- **E. Monzio Compagnoni**, R. Islamov, F. N. Proske, E. Gorbunov, A. Lucchi, A. Orvieto. *On the Interaction of Batch Noise, Adaptivity, and Compression under (L_0, L_1) -Smoothness: An SDE Approach*. **ICML**, 2026.
- **E. Monzio Compagnoni**, A. Stanghellini, R. Islamov, A. Lucchi, A. Koloskova. *Adaptive Methods Are Preferable in High Privacy Settings: An SDE Perspective*. **ICLR**, 2026.
- **E. Monzio Compagnoni**, T. Liu, R. Islamov, F. N. Proske, A. Orvieto, A. Lucchi. *Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise*. **ICLR**, 2025.
- **E. Monzio Compagnoni**, R. Islamov, F. N. Proske, A. Lucchi. *Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs*. **AISTATS (Oral)**, 2025.
- **E. Monzio Compagnoni**, A. Orvieto, H. Kersting, F. N. Proske, A. Lucchi. *SDEs for Minimax Optimization*. **AISTATS**, 2024.
- **E. Monzio Compagnoni**, L. Biggio, A. Orvieto, F. N. Proske, H. Kersting, A. Lucchi. *An SDE for Modeling SAM: Theory and Insights*. **ICML**, 2023.
- **E. Monzio Compagnoni**, A. Scampicchio, L. Biggio, A. Orvieto, T. Hofmann, J. Teichmann. *On the Effectiveness of Randomized Signatures as Reservoir for Learning Rough Dynamics*. **IJCNN**, 2023.

Other Publications

- C. Zhang, R. Islamov, **E. Monzio Compagnoni**, J. Pang, A. Lucchi, A. Orvieto. *Beyond a Single Explanation of the Adam--SGD Gap*. **arXiv preprint**, 2026.
- M. Forzo, A. Russo, **E. Monzio Compagnoni**, A. Pacchiano. *A Diffusion Approximation for Temporal-Difference Learning with Linear Features under Markovian Noise*. **ICML Workshop (DEMO)**, 2026.
- S. S. Ramesh, R. Faro, **E. Monzio Compagnoni**, I. Bogunovic, A. Lucchi. *Frontier Learning: Training LLM Reasoners at the Edge of Capability*. **ICML Workshop (FoGen)**, 2026.
- M. Burzoni, A. Doldi, **E. Monzio Compagnoni**. *Risk Sharing with Deep Neural Networks*. **Quantitative Finance**, 2024.

References

- PhD Supervisor: Prof. Dr. Aurelien Lucchi: aurelien.lucchi@unibas.ch
- External Collaborator: Dr. Antonio Orvieto: antonio@tue.ellis.eu